

Equilibrium of Forces

The meaning of equilibrium · $\Sigma F = 0$ on each axis · resolving forces · cables, ropes & springs

Physics — Grade 11
Explanation &
Questions
 $g = 9.8 \text{ m/s}^2$

1 What does equilibrium mean?

A body is in **equilibrium** when the resultant of all the forces acting on it is **zero**. Its velocity then does not change, so the body either:

- stays **at rest**, or
- moves in a straight line with a **constant velocity**.

$$\Sigma F = 0$$

the condition of equilibrium : the vector sum of all the forces is zero

Remember

Equilibrium does **not** mean “no forces”. Several forces may act on the body, but they cancel each other, so their **resultant** is zero.

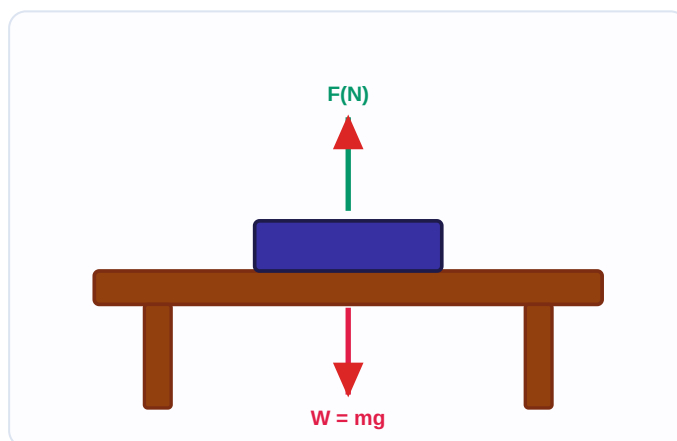


Fig. 1 — a book resting on a table.

A book resting on a table is in equilibrium under two forces (Fig. 1):

Force	Direction	Source
the weight $W = m g$	vertically downwards	the pull of the Earth on the book
the normal force F_N	vertically upwards	the push of the table on the book (a contact force)

Since the book is at rest, $F_N = W$. If a second book is placed on top, the table must support both, so the normal force on the lower book becomes $2 m g$, while the weight of the lower book itself stays $m g$.

2 Applying $\Sigma F = 0$ on each axis

$\Sigma F = 0$ is a vector equation. We apply it to each axis **separately**: the forces along the horizontal (x) axis must cancel, and the forces along the vertical (y) axis must cancel.

$$\Sigma F_x = 0 \quad \text{and} \quad \Sigma F_y = 0$$

forces to the right = forces to the left | forces upwards = forces downwards

A force that acts at an angle is first **resolved** into two perpendicular components. For a force F making an angle θ with the **horizontal**:

$$F_x = F \cos \theta \quad F_y = F \sin \theta$$

if the angle is measured from the **vertical**, the sin and the cos change places

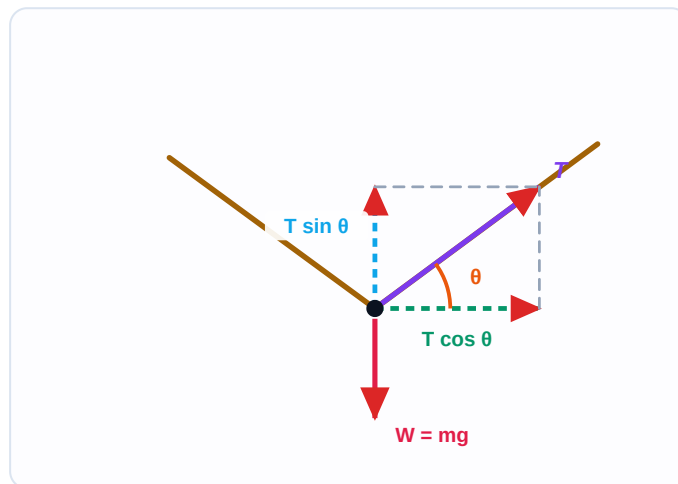


Fig. 2 — a tension T resolved into a horizontal and a vertical component.

3 Forces along one straight line

When all the forces act along the same vertical line, the forces upwards must equal the forces downwards. The missing force is the one that makes the sum zero; its sign tells us its direction (Fig. 3).

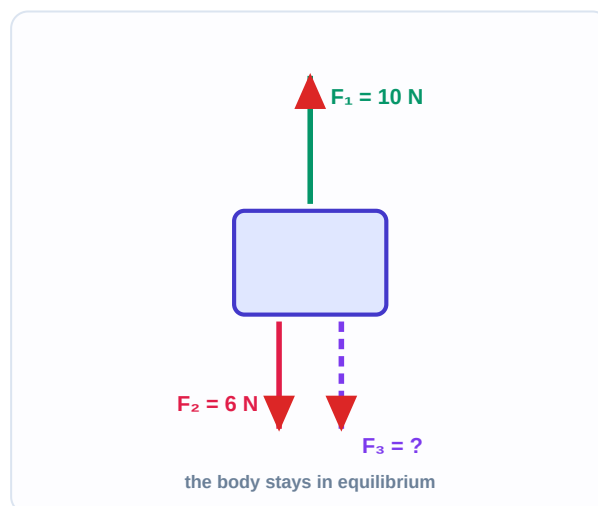


Fig. 3 — three forces along one vertical line.

4 A weight hanging from two identical cables

A lamp of weight W hangs from two identical cables, each making the same angle α with the **horizontal** (Fig. 4). By symmetry the two tensions are equal:

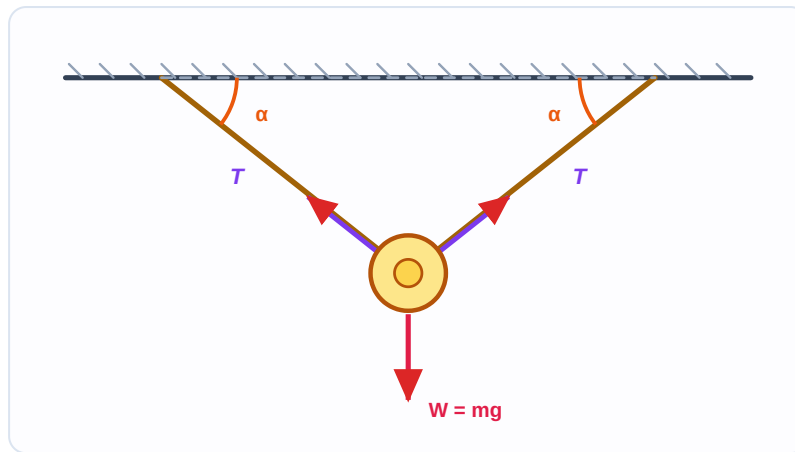


Fig. 4 — a lamp supported by two symmetric cables.

- **Horizontal axis:** the two horizontal components $T \cos \alpha$ are equal and opposite, so they cancel each other.
- **Vertical axis:** the two vertical components together carry the weight.

$$2 T \sin \alpha = W \Rightarrow T = W / (2 \sin \alpha)$$

α = the angle between each cable and the horizontal

α (with the horizontal)	90°	60°	30°	10°	5°
T / W	0.5	0.58	1	2.9	5.7

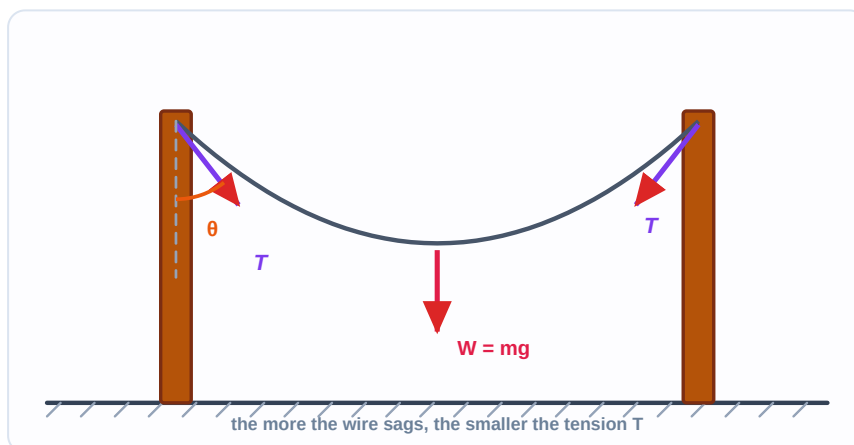


Fig. 5 — a wire fixed between two poles carrying a weight at its middle.

The sagging wire

If the wire sags more, the angle α with the horizontal increases, $\sin \alpha$ increases and the tension **decreases**. If the wire is pulled flatter, $\alpha \rightarrow 0$ and $T = W / (2 \sin \alpha)$ becomes very large. That is why a wire carrying a weight can never be made perfectly horizontal.

5 A lamp held by a cable and a horizontal rope

A lamp hangs from a cable fixed to the ceiling and is held aside by a light horizontal rope fixed to a wall, so that the cable makes an angle θ with the **vertical** (Fig. 6). Three forces act on the lamp: the tension T in the cable, the pull P of the rope and the weight W .

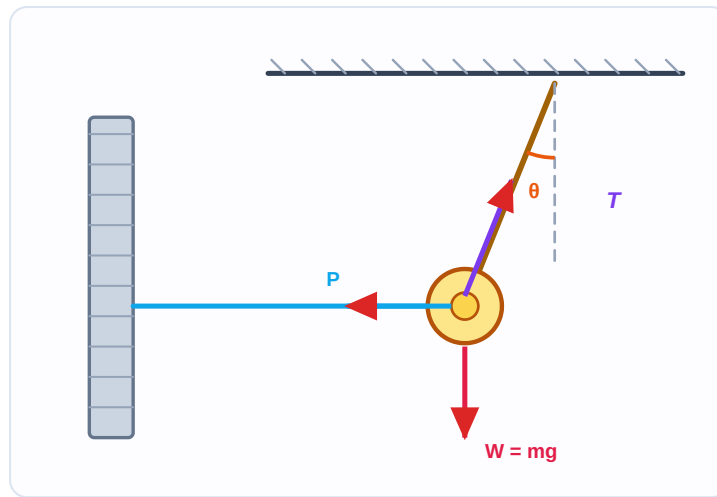


Fig. 6 — three forces keep the lamp in equilibrium.

$$\text{vertical : } T \cos \theta = W \quad \text{horizontal : } T \sin \theta = P$$

$$\theta \text{ with the vertical} \mid T = \sqrt{W^2 + P^2} \mid \tan \theta = P / W$$

Which component carries the weight?

Only the **vertical** component of the tension appears in the vertical equation. The rope is horizontal, so it has no vertical component and is balanced by the horizontal component of the tension alone.

6 Two cables making different angles

If a sign hangs from two cables making **different** angles α_1 and α_2 with the horizontal (Fig. 7), the two tensions are **not** equal. We write the two equations and solve them together:

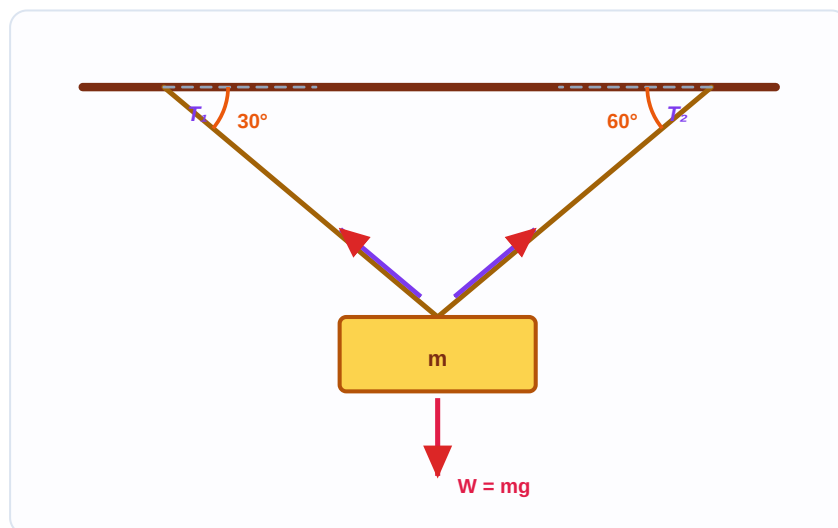


Fig. 7 — a sign supported by two cables at different angles.

$$\text{horizontal : } T_1 \cos \alpha_1 = T_2 \cos \alpha_2$$

$$\text{vertical : } T_1 \sin \alpha_1 + T_2 \sin \alpha_2 = W$$

The cable that is **closer to the vertical** (the larger angle with the horizontal) carries the **larger** tension.

7 A body hanging from two identical springs

A body hangs at rest from two identical light springs, each making an angle θ with the **vertical** (Fig. 8). Each spring pulls with the same force F , and the two vertical components carry the weight:

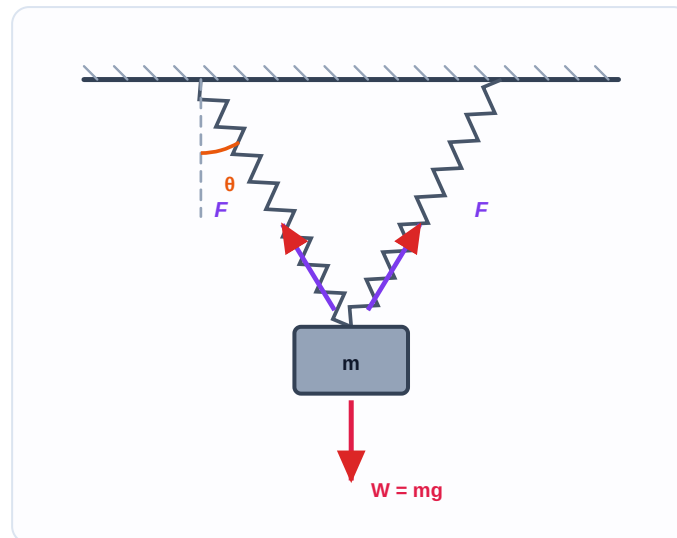


Fig. 8 — a body hanging from two identical springs.

$$2 F \cos \theta = W \quad \text{and} \quad F = k x$$

θ with the **vertical** | k = spring constant (N/m) | x = extension of each spring

8 How to solve an equilibrium problem

1. Draw the body alone and mark **every** force acting on it (weight, tensions, normal force, pulls).
2. Change the mass into a weight : $W = m g$.
3. Resolve every inclined force into a horizontal and a vertical component.
4. Write $\Sigma F_x = 0$ and $\Sigma F_y = 0$, and solve.

★ Key facts you must memorise

- Equilibrium : the body is at rest or moves with a constant velocity $\Rightarrow \Sigma \mathbf{F} = \mathbf{0}$.
- Apply the condition to each axis : $\Sigma F_x = 0$ and $\Sigma F_y = 0$.
- Components : $F_x = F \cos \theta$, $F_y = F \sin \theta$ (θ from the horizontal).
- Two symmetric cables : $T = W / (2 \sin \alpha)$; the flatter the cables, the larger the tension.
- Cable + horizontal rope : $T \cos \theta = W$ and $T \sin \theta = P$ (θ from the vertical).
- A book on a table : $F_N = W$; the weight comes from the Earth, the normal force from the table.

Common mistakes

- Using the mass in kilograms as a force : always write $W = m g$ first.
- Using $\sin$ instead of $\cos$: check whether θ is measured from the horizontal or from the vertical.
- Writing $T = W$ for a cable that is not vertical.
- Adding a horizontal force into the vertical equation.

9 Solved examples

Example 1 — a book on a table

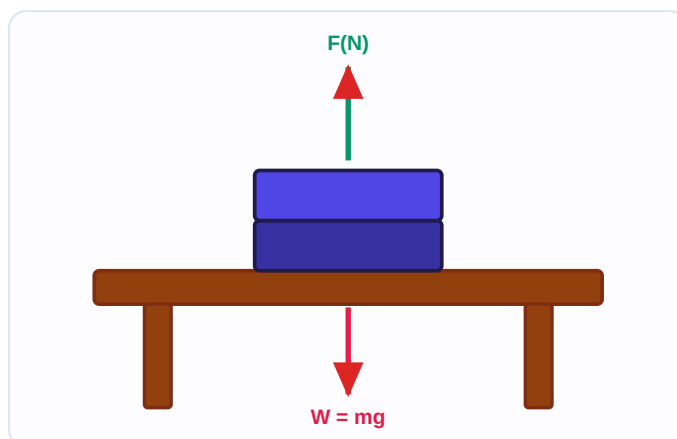


Fig. 9 — two identical books on a table.

Given: A book of mass 0.5 kg rests on a table. Find the normal force on it. A second identical book is then placed on top of it. Find the new normal force that the table exerts on the lower book, and the weight of the lower book.

Solution

one book : $F(N) = W = m g = 0.5 \times 9.8 = 4.9 \text{ N}$

two books : the table supports both books : $F(N) = 2 m g = 2 \times 4.9 = 9.8 \text{ N}$

the weight of the lower book depends only on its own mass : $W = 4.9 \text{ N}$
(unchanged)

$F(N) : 4.9 \text{ N} \rightarrow 9.8 \text{ N}$; the weight stays 4.9 N

Example 2 — the third force along one line

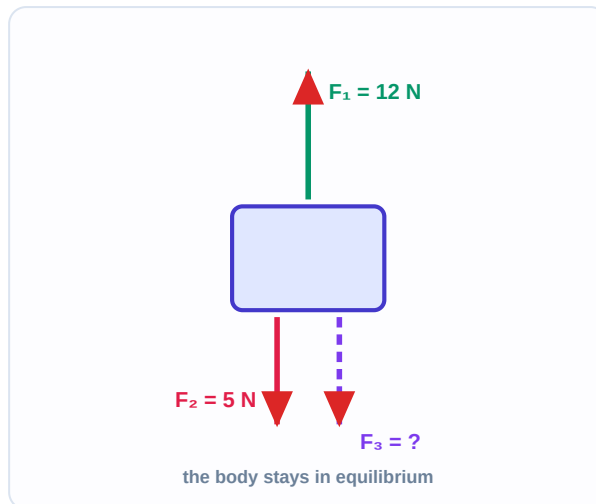


Fig. 10 — find the third force.

Given: A body is in equilibrium under three forces along one vertical line : 12 N upwards, 5 N downwards and a third force F_3 . Find F_3 .

Solution

$$\begin{aligned} \text{take upwards as positive : } \quad \Sigma F(y) &= 0 \\ + 12 - 5 + F_3 &= 0 \quad \Rightarrow \quad F_3 = - 7 \text{ N} \end{aligned}$$

The minus sign means that F_3 points **downwards**.

$F_3 = 7 \text{ N downwards}$

Example 3 — a lamp on two symmetric cables

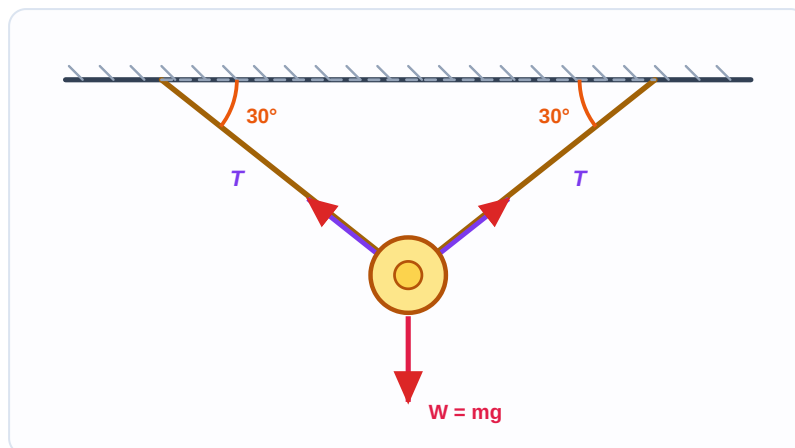


Fig. 11 — each cable makes 30° with the horizontal.

Given: A lamp of mass 0.6 kg hangs from two identical cables, each making 30° with the horizontal. Find the tension in each cable.

Solution

$$W = m g = 0.6 \times 9.8 = 5.88 \text{ N}$$

horizontal : the two components $T \cos 30^\circ$ cancel

$$\text{vertical : } 2 T \sin 30^\circ = W \Rightarrow 2 T \times 0.5 = 5.88 \Rightarrow T = 5.88 \text{ N}$$

At 30° each cable carries a tension **equal to the whole weight**.

$$\boxed{T = 5.88 \text{ N}}$$

Example 4 — a cable and a horizontal rope

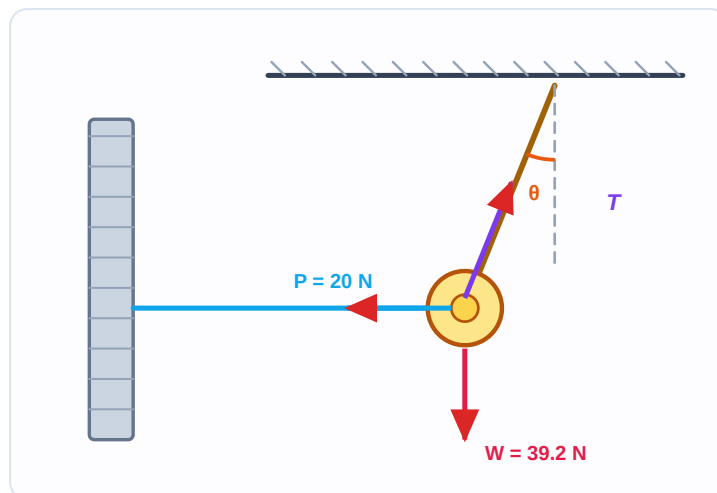


Fig. 12 — the rope pulls the lamp with $P = 20 \text{ N}$.

Given: A lamp of mass 4.0 kg hangs from a cable and is held aside by a horizontal rope whose tension is 20 N . Find the tension T in the cable and the angle θ that the cable makes with the vertical.

Solution

$$W = m g = 4.0 \times 9.8 = 39.2 \text{ N}$$

$$\text{vertical : } T \cos \theta = W = 39.2 \text{ N}$$

$$\text{horizontal : } T \sin \theta = P = 20 \text{ N}$$

$$T = \sqrt{(39.2^2 + 20^2)} = \sqrt{1936.6} = 44.0 \text{ N}$$

$$\tan \theta = P / W = 20 / 39.2 = 0.51 \quad \Rightarrow \quad \theta = 27^\circ$$

$$T = 44.0 \text{ N}, \theta = 27^\circ \text{ with the vertical}$$

Example 5 — two cables at different angles

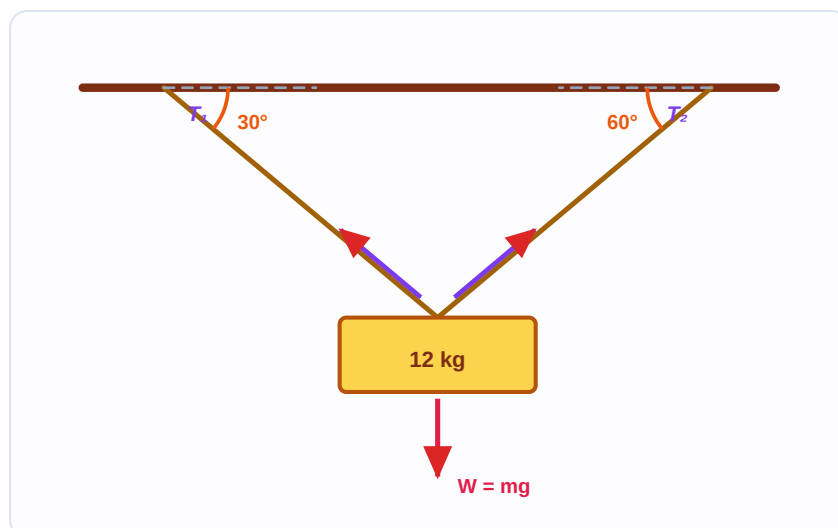


Fig. 13 — a sign of mass 12 kg on two cables.

Given: A sign of mass 12 kg hangs from two cables making 30° and 60° with the horizontal. Find the tension in each cable. ($\sin 30^\circ = \cos 60^\circ = 0.5$, $\cos 30^\circ = \sin 60^\circ = 0.866$)

Solution

$$W = m g = 12 \times 9.8 = 117.6 \text{ N}$$

$$\text{horizontal : } T_1 \cos 30^\circ = T_2 \cos 60^\circ \Rightarrow 0.866 T_1 = 0.5 T_2 \Rightarrow T_2 = 1.732 T_1$$

$$\text{vertical : } T_1 \sin 30^\circ + T_2 \sin 60^\circ = 117.6$$

$$0.5 T_1 + (1.732 T_1)(0.866) = 117.6 \Rightarrow 2 T_1 = 117.6$$

$$T_1 = 58.8 \text{ N}, \quad T_2 = 1.732 \times 58.8 = 101.8 \text{ N}$$

The steeper cable (60°) carries the larger tension.

$$T_1 = 58.8 \text{ N}, T_2 = 101.8 \text{ N}$$

Example 6 — two identical springs

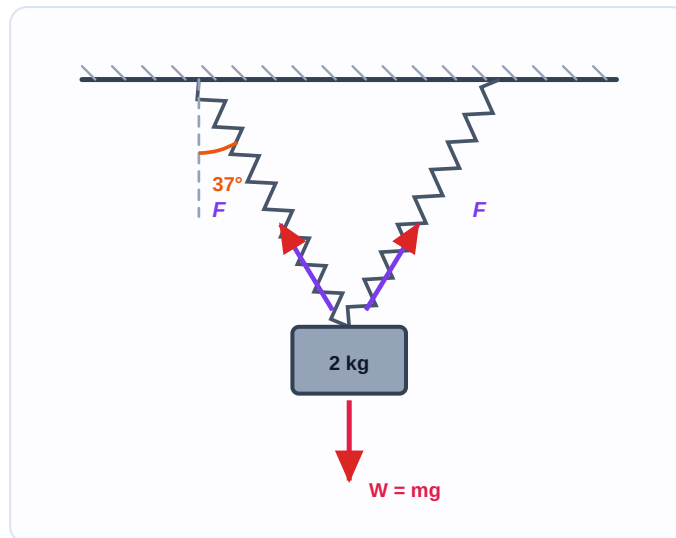


Fig. 14 — each spring makes 37° with the vertical.

Given: A body of mass 2 kg hangs at rest from two identical springs, each making 37° with the vertical. The spring constant of each spring is 100 N/m. Find the force in each spring and its extension. ($\cos 37^\circ = 0.8$)

Solution

$$W = m g = 2 \times 9.8 = 19.6 \text{ N}$$

$$\text{vertical : } 2 F \cos 37^\circ = W \Rightarrow 2 F \times 0.8 = 19.6 \Rightarrow F = 12.25 \text{ N}$$

$$\text{extension : } x = F / k = 12.25 / 100 = 0.1225 \text{ m}$$

$$F = 12.25 \text{ N , } x = 12.25 \text{ cm}$$

10 Exercises

20 questions. Take $g = 9.8 \text{ m/s}^2$, $\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.866$, $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$, $\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$, $\sin 60^\circ = 0.866$, $\cos 60^\circ = 0.5$. Show every step and give every answer with its unit.

Q1 A body is in a state of equilibrium when:

MCQ

- ☐ A) its velocity is always zero
- ☐ B) it stays at rest or moves with a constant velocity
- ☐ C) its velocity changes continuously
- ☐ D) no forces act on it

Q2 The condition of equilibrium of a body is written mathematically as:

MCQ

- ☐ A) $\Sigma F = m g$
- ☐ B) $\Sigma F = 0$
- ☐ C) $\Sigma F = m a$, where $a \neq 0$
- ☐ D) $\Sigma F = F_1$ only

Q3 A car moves along a straight road with a **constant velocity**. The resultant force acting on it is:

MCQ

- ☐ A) in the direction of motion
- ☐ B) zero
- ☐ C) opposite to the motion
- ☐ D) equal to its weight

Q4 A body is in equilibrium under three forces along one vertical line: 15 N upwards, 9 N downwards and a third force. The third force is:

MCQ

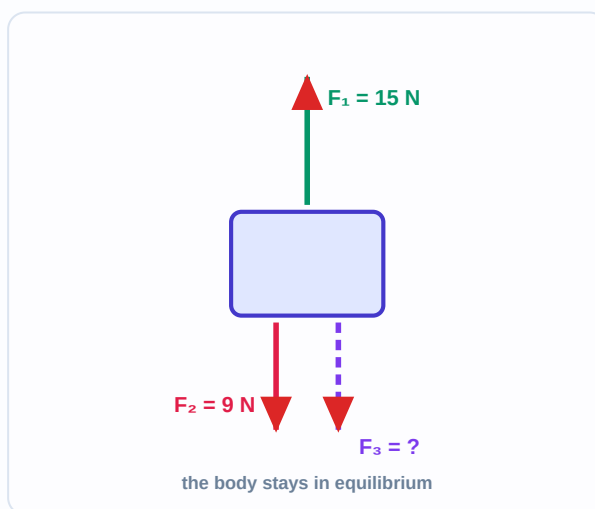


Fig. 15

- ☐ A) 6 N upwards
- ☐ B) 6 N downwards
- ☐ C) 24 N upwards
- ☐ D) zero

Q5 A book of mass 1.2 kg rests on a horizontal table. The normal force that the table exerts on the book is:

MCQ

A) 1.2 N

B) 9.8 N

C) 11.76 N

D) zero

Q6 For the book of Q5, a second identical book is placed on top of it. **State** which force on the lower book changes and which stays the same, and give the **source** of each force.

Explain

Q7 A lamp of mass 2 kg hangs from two identical cables, each making 30° with the horizontal. The tension in each cable is:

MCQ

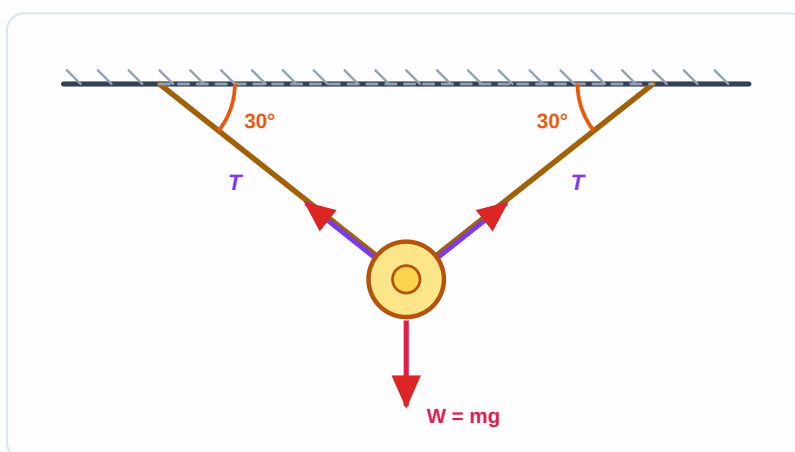


Fig. 16

A) 9.8 N

B) 39.2 N

C) 11.3 N

D) 19.6 N

Q8 For the lamp of Q7, if the two cables are made **flatter** (the angle with the horizontal decreases) while the load stays the same, the tension in each cable:

MCQ

A) increases

B) decreases

C) stays the same

D) becomes zero

Q9 A sign of mass 5 kg hangs from two identical cables, each making 60° with the horizontal. Find the tension in each cable.

Problem

Q10 A wire fixed between two poles carries a weight at its middle. **Explain** what happens to the tension in the wire if it is allowed to sag more, and why it can never be made perfectly horizontal.

Explain

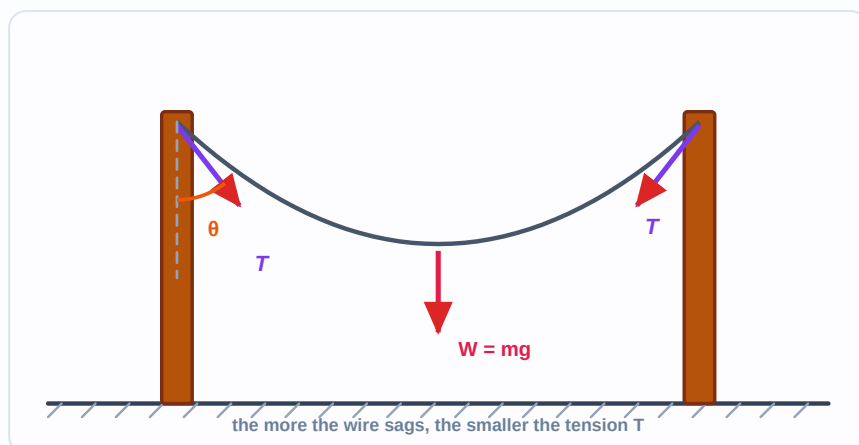


Fig. 17

Q11 A lamp hangs from a cable fixed to the ceiling and is held aside by a horizontal rope. The **vertical** component of the tension in the cable is equal to:

MCQ

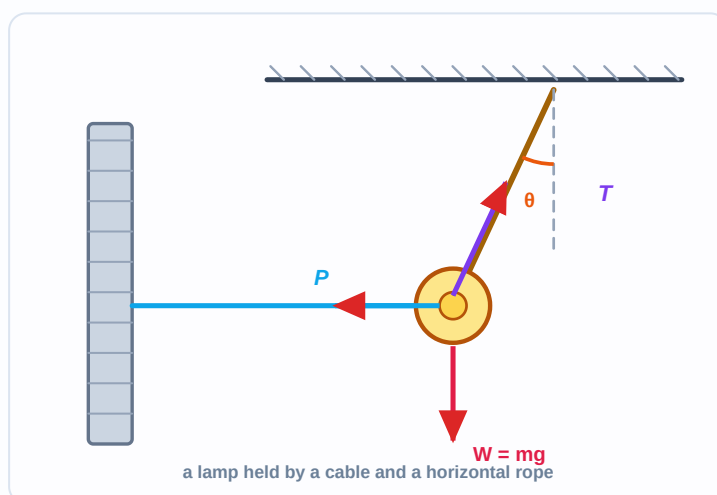


Fig. 18

A) zero

B) the weight of the lamp

C) the tension in the horizontal rope

D) half the weight of the lamp

Q12 A lamp of mass 3 kg is held aside by a horizontal rope, so that its cable makes 37° with the vertical. Find the tension in the cable and the tension in the rope.

Problem

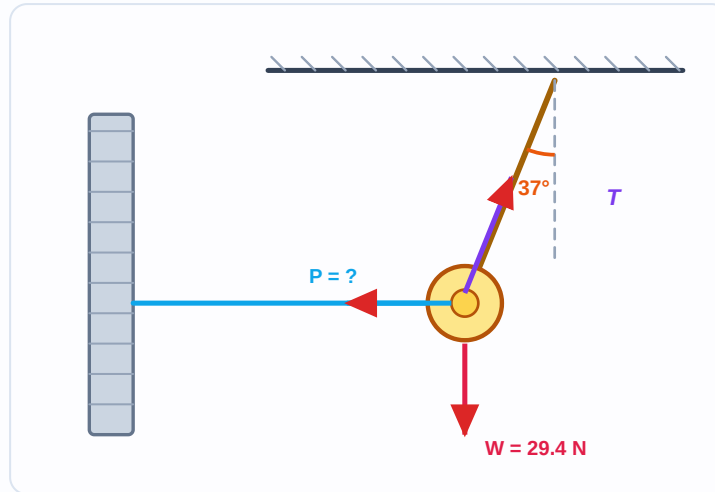


Fig. 19

Q13 A lamp of weight 24 N is held aside by a horizontal rope whose tension is 7 N. Find the tension in the cable and the angle it makes with the vertical.

Problem

Q14 For a lamp hanging in equilibrium from two identical cables making the same angle with the horizontal, the two horizontal components of the tensions:

MCQ

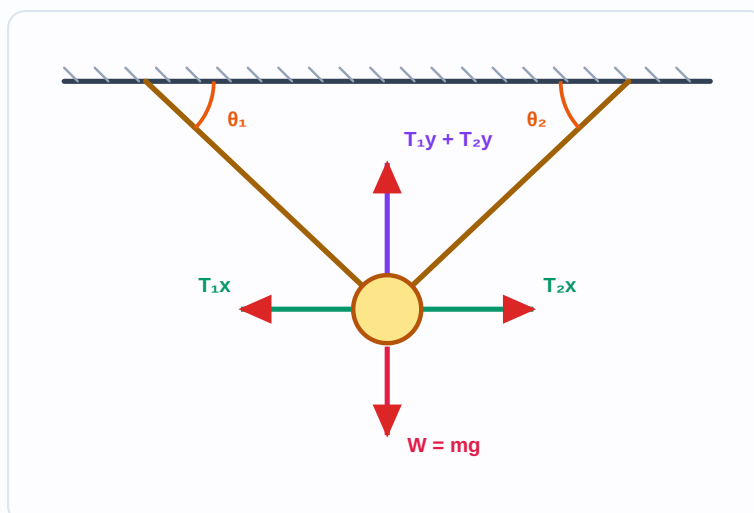


Fig. 20

A) are equal and opposite, so they cancel each other

B) add together

C) balance the weight

D) are each equal to the weight

Q15 A sign hangs from two cables making **different** angles with the horizontal. The larger tension is in:

MCQ

A) the flatter cable

B) neither — the two tensions are always equal

C) the steeper cable (closer to the vertical)

D) the cable fixed farther from the sign

Q16 A sign of mass 10 kg hangs from two cables making 37° and 53° with the horizontal. Find the tension in each cable.

Problem

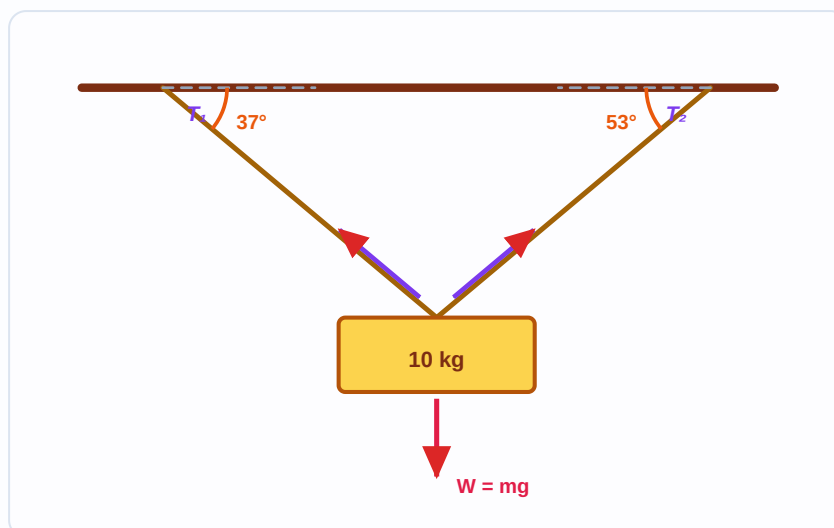


Fig. 21

Q17 A body of mass 2.5 kg hangs at rest from two identical springs, each making 37° with the vertical. The weight of the body is:

MCQ

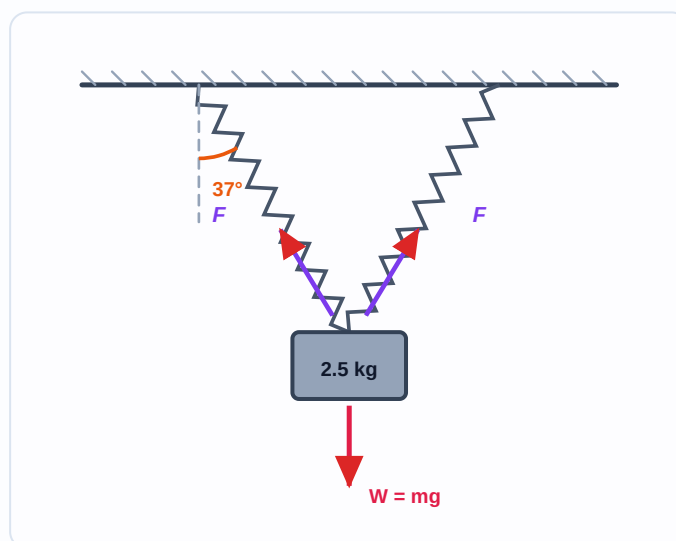


Fig. 22

A) 2.5 N

B) 24.5 N

C) 12.25 N

D) 49 N

Q18 For the body of Q17, the spring constant of each spring is 50 N/m. Find the force in each spring and its extension.

Problem

Q19 Two identical springs hold the same body. If the angle that each spring makes with the **vertical** is increased, the force in each spring:

MCQ

A) increases

B) decreases

C) stays the same

D) becomes zero

Q20 A lamp hangs from two cables making 30° and 60° with the horizontal. The tension in the 30° cable is 20 N. Find the tension in the other cable and the weight of the lamp.

★ HOTS

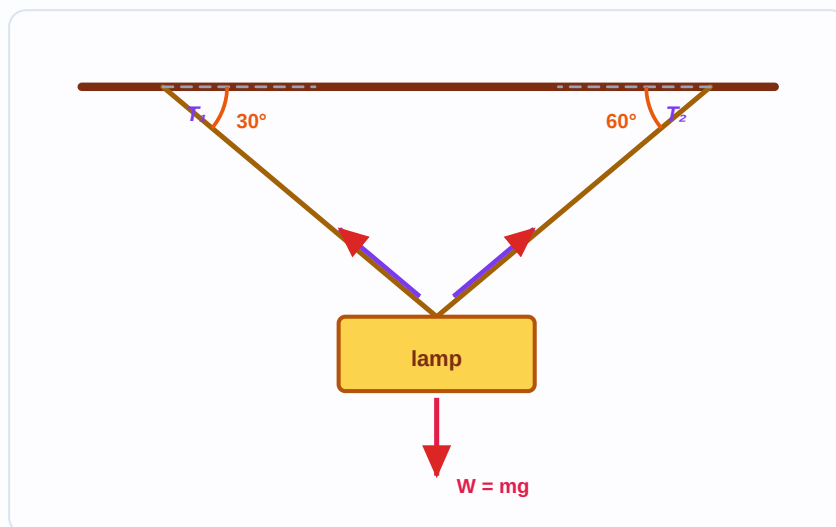


Fig. 23

11 Model answer — final answers only

Use this page to correct the exercises. Only the final answer is given, so the student must write every step of the working.

Q	Final answer	Q	Final answer
1	B) it stays at rest or moves with a constant velocity	2	B) $\Sigma F = 0$
3	B) zero	4	B) 6 N downwards
5	C) 11.76 N	6	F(N) doubles to 23.52 N (from the table) ; the weight stays 11.76 N (from the Earth)
7	D) 19.6 N	8	A) increases
9	T = 28.3 N	10	more sag $\rightarrow$ larger angle with the horizontal $\rightarrow$ smaller tension ; a flat wire needs an infinite tension
11	B) the weight of the lamp	12	T = 36.75 N , P = 22.05 N
13	T = 25 N , $\theta = 16.3^\circ$ with the vertical	14	A) are equal and opposite, so they cancel each other
15	C) the steeper cable (closer to the vertical)	16	T(37°) = 58.8 N , T(53°) = 78.4 N
17	B) 24.5 N	18	F = 15.3 N , x = 30.6 cm
19	A) increases	20	T(60°) = 34.6 N , W = 40 N